

EMCS-NEC-PUR-012-02

EMCS Design Review Meeting DCN

26 September 2023

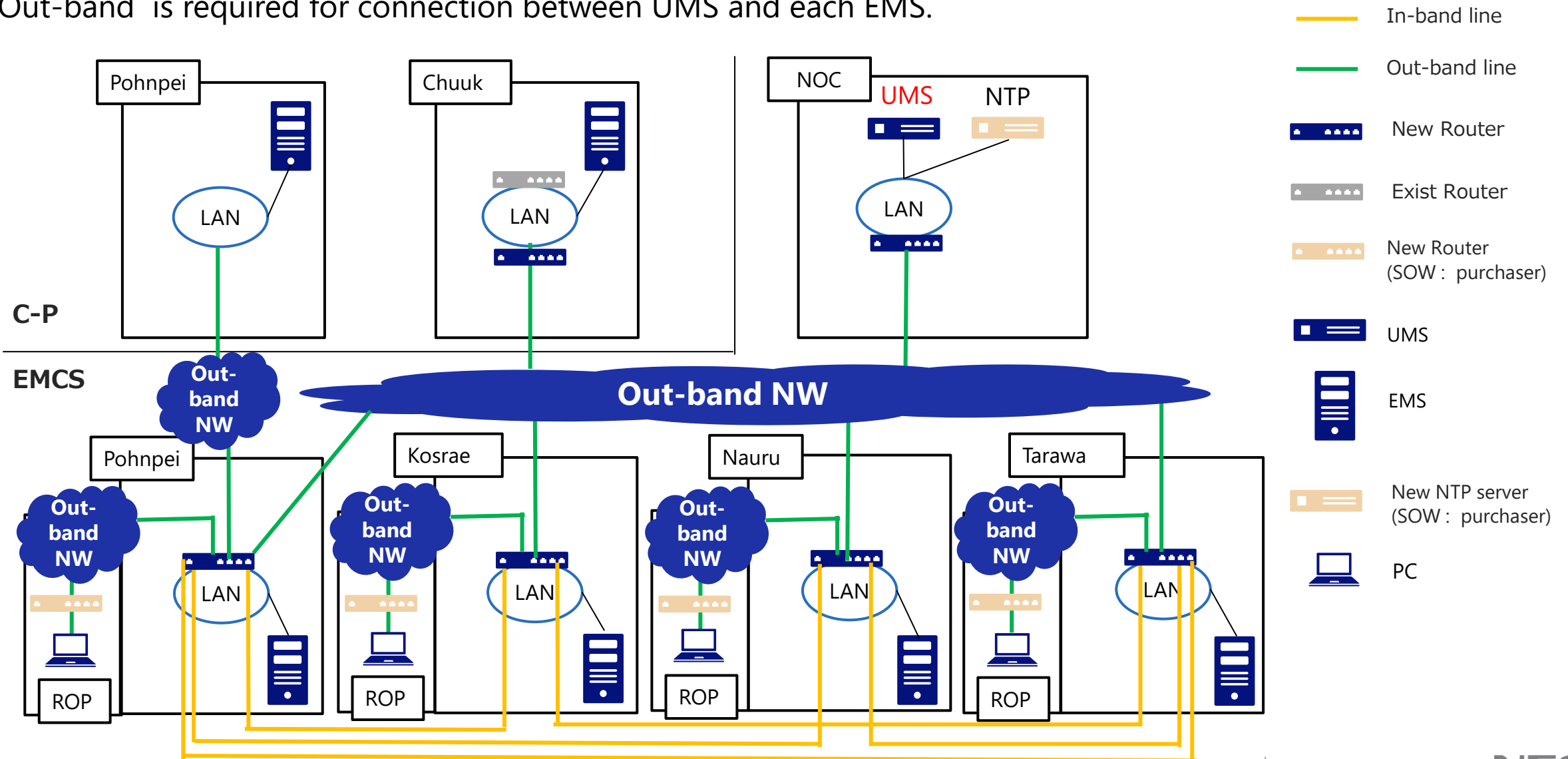
NEC Corporation

Content

1. DCN image
2. DCN policy plan
3. DCN SOW
4. Proposed DCN LAN Network Design
5. DCN Manager
6. Request to purchaser

DCN image

Out-band is required for connection between UMS and each EMS.



DCN policy plan

EMCS DCN plan is as follows.

No	Contents
1	DCN WAN uses out-band and in-band
2	Routing between NOC and CLSs uses OSPF. Routing between ROP and CLS uses static.
3	If the connection between CLS and ROP uses the Internet, it will be encrypted with IPsec on the router.
4	EMCS CLSs(Tarawa, Kosrae Nauru and new Pohnpei) have router×2(spear×1) and L2switch×2.(spear×1) .
5	Exist Chuuk has router×1(spear×1) and L2switch×1(spear×1) .
6	Exist Pohnpei has router×1(spear×1) and L2switch×1.
7	ROPs have router(purchaser SOW)×1.
8	NOC has router×1(spear×1) and L2switch×1(spear×1).
9	NTP server(purchaser SOW)× is placed in NOC.
10	IP phones are placed in NOC and EMCS CLSs(Tarawa, Kosrae Nauru and new Pohnpei) .

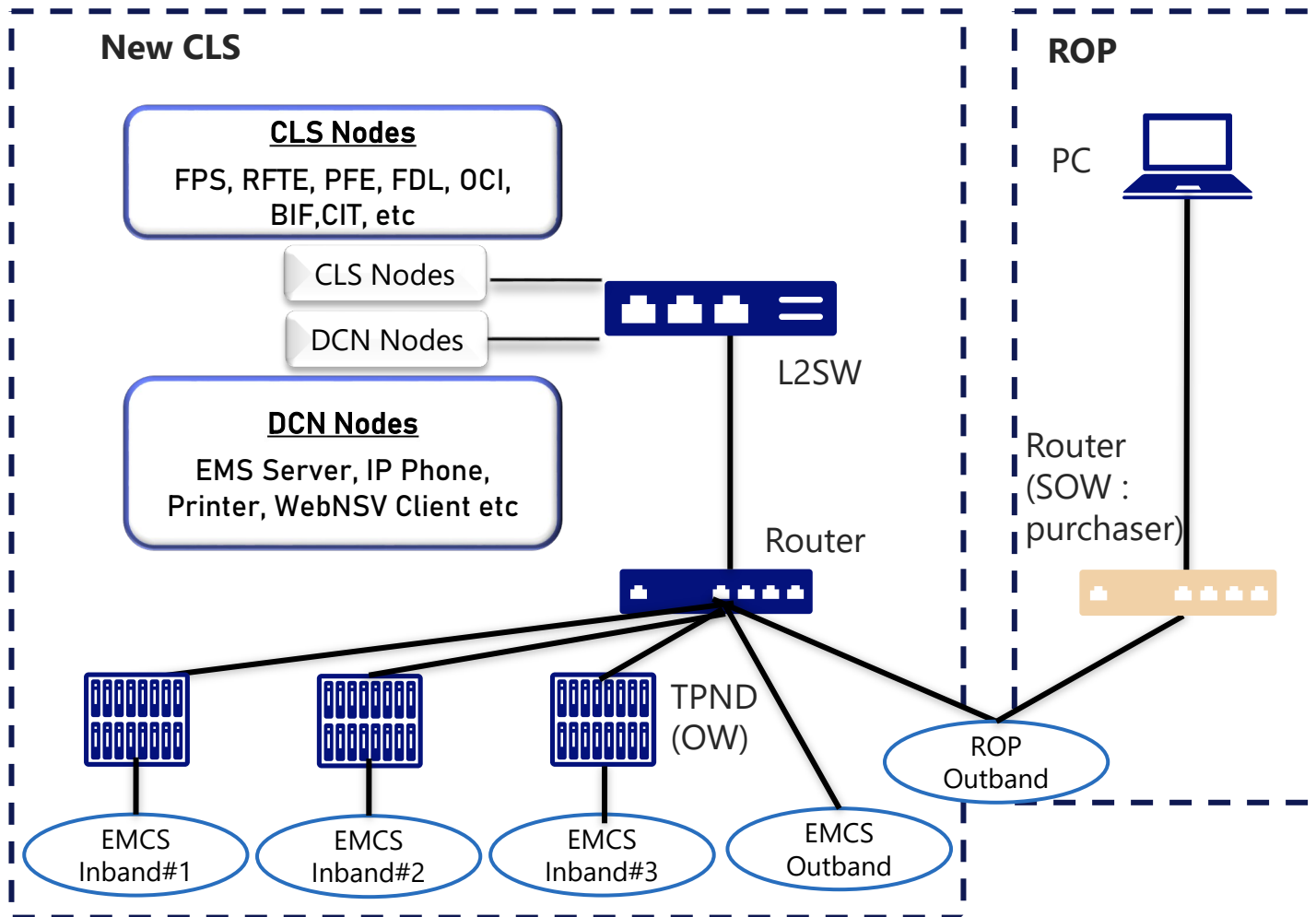
DCN policy plan

EMCS DCN plan is as follows.

No	EMCS CLS		C-P Chuuk CLS		C-P Pohnpei CLS		NOC		ROP	
	Purchaser	NEC	Purchaser	NEC	Purchaser	NEC	Purchaser	NEC	Purchaser	NEC
Out-band network (NOC to CLS)	○		○		○		○		N/A	N/A
Out-band network (ROP to CLS)	○		N/A	N/A	N/A	N/A	N/A	N/A	○	
Out-band network (Exist Pohnpei to New Pohnpei CLS)	○		N/A	N/A	○		N/A	N/A	N/A	N/A
Router		○ 1(1)		○ 1(1)				○ 1(1)	○ 1	
L2 Switch		○ 1(1)		○ 1(1)				○ 1(1)	○ 1	
IP Phone		○ 1(1)		○ 1(1)				○ 1(1)	N/A	N/A
DCN manager								○ 1(1)		
NTP server							○ 1			

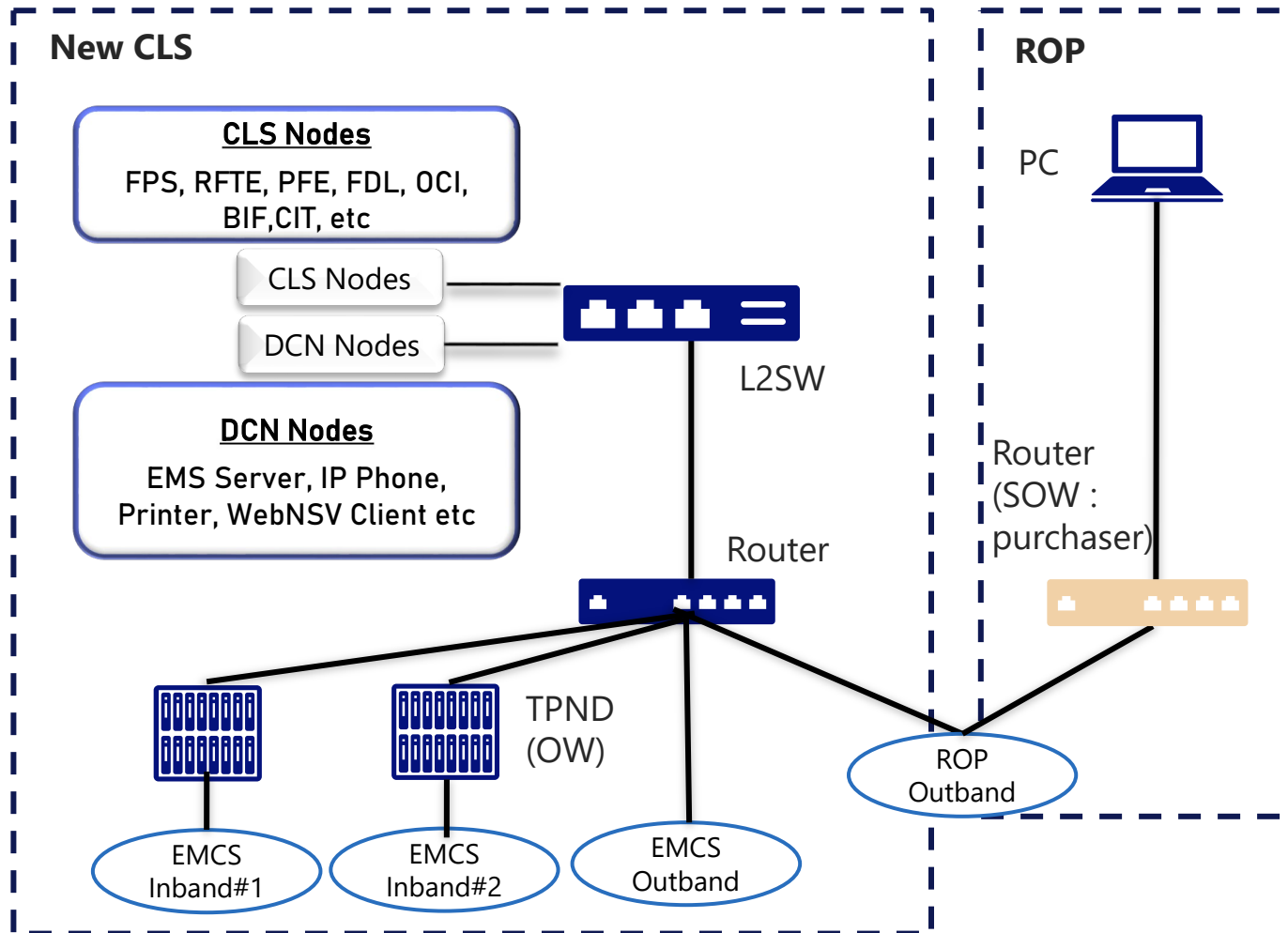
Proposed DCN LAN Network Design (1/5)

Tarawa DCN LAN



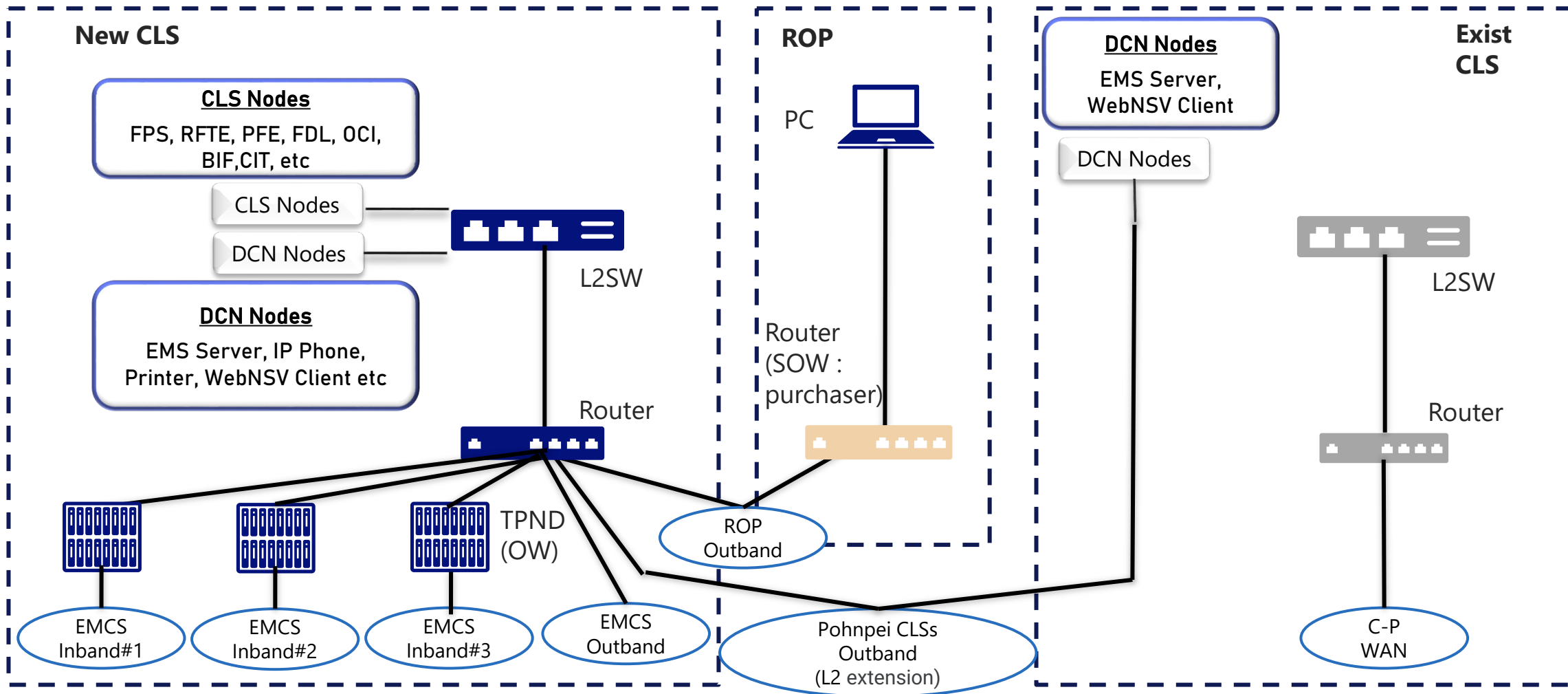
Proposed DCN LAN Network Design (2/5)

Kosrae and Nauru DCN LAN



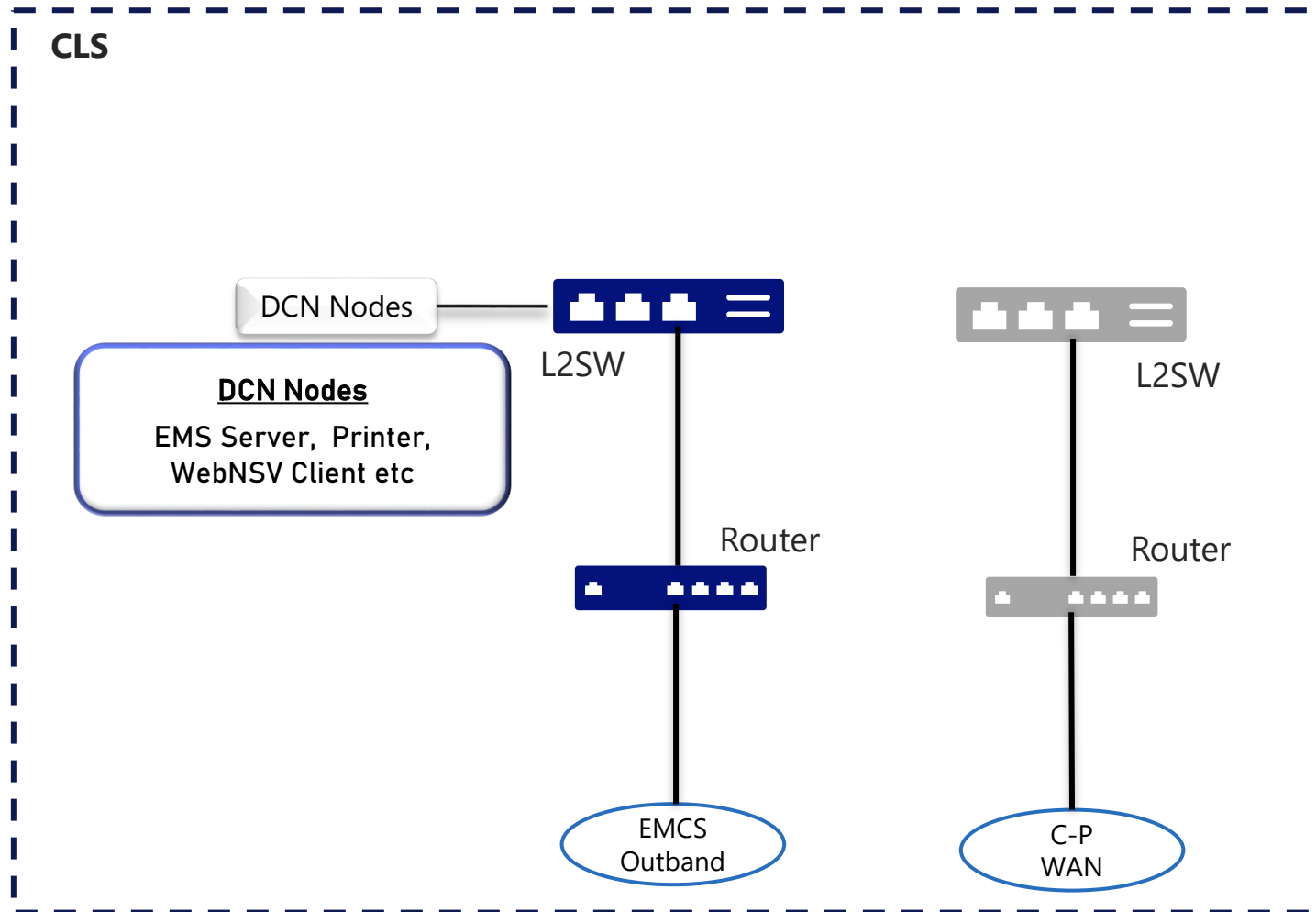
Proposed DCN LAN Network Design (3/5)

Pohnpei DCN LAN



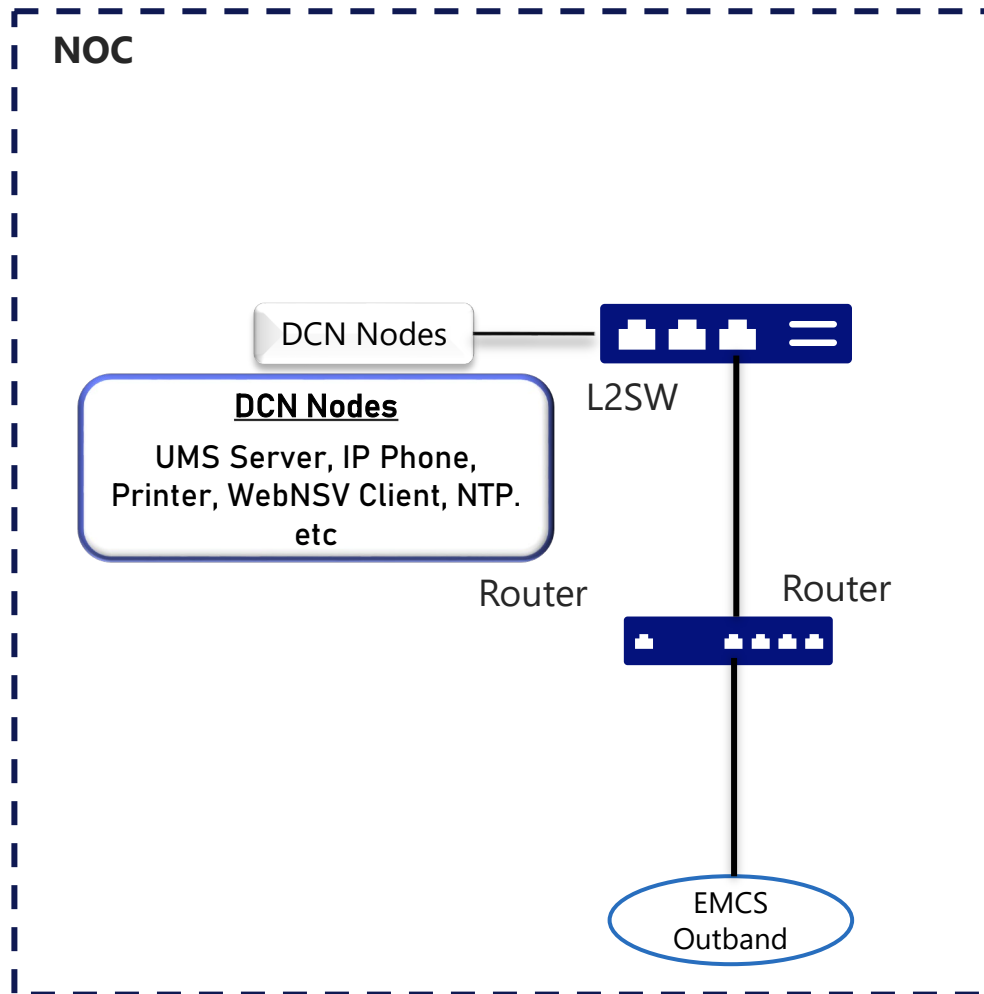
Proposed DCN LAN Network Design (4/5)

Chuuk DCN LAN



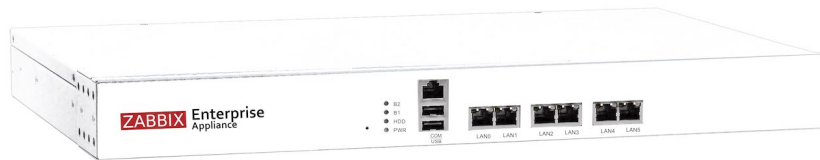
Proposed DCN LAN Network Design (5/5)

NOC DCN LAN



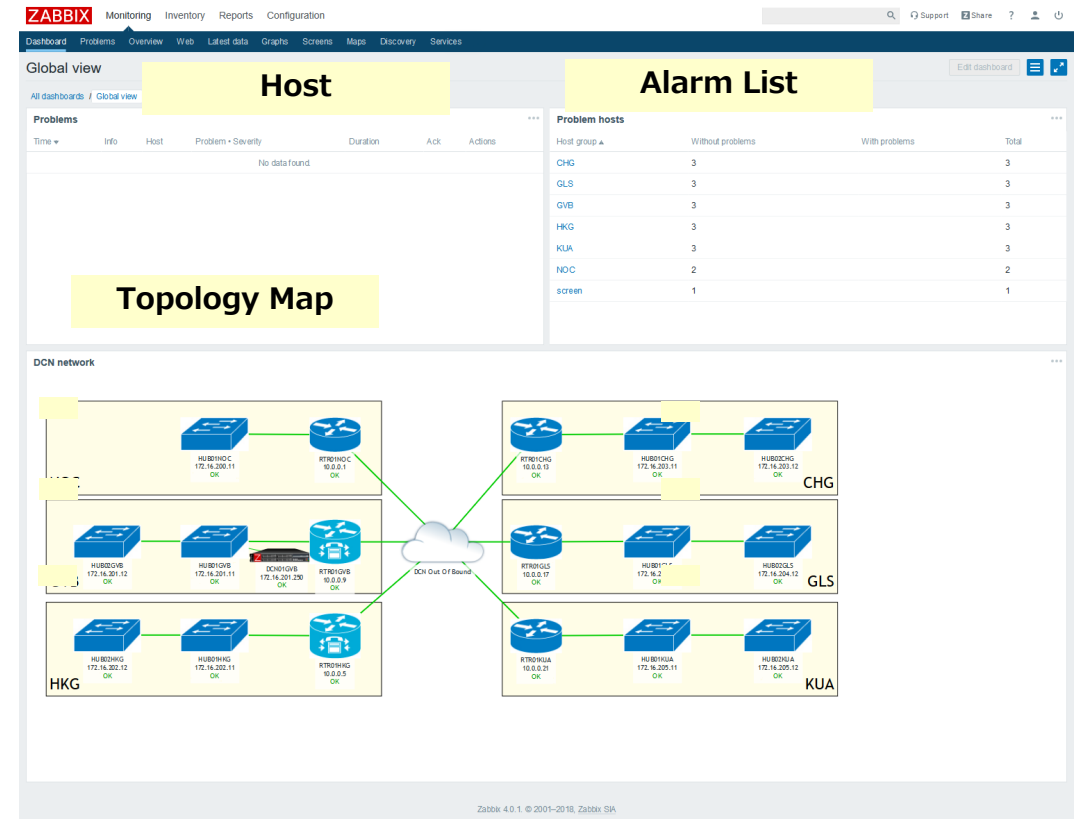
DCN Manager

Zabbix Enterprise Appliance ZX-5400



- DCN Equipment Monitoring
- Fail Notification
- Power: AC Power

Sample of Zabbix Web GUI



Request to purchaser

No	Contents
1	Provision of out-of-band network that meet the following specifications Bandwidth: NOC 10Mbps or more CLS 3Mbps or more. Latency Recommended within 100msec Allowable within 200msec
2	Provision of ROP network that meet the following specifications Bandwidth: 1Mbps or more.
3	Providing a line between the existing Pohnpei and the new Pohnpei that meets the following specifications. Bandwidth: 10Mbps or more.
4	Provision and installation of NTP server.
5	Provision and installation of ROP routers

\Orchestrating a brighter world

NEC